Single-Task and Multi-Task Learning with A Century of Portraits

Machine Learning Mid-Term Project

1. Introduction

This project compares year regression from American high-school portraits with a multi-task network that jointly predicts year and gender. Gender was selected because it is directly encoded by the F/M folders, has a reliable binary target, and can regularize shared visual features while remaining substantially easier than noisy multi-class state prediction.

2. Dataset and preprocessing

The official archive yielded 29,152 training-list entries and 7,408 test-list entries (36,560 unique referenced images), spanning 1905–2013. A seed-1 random 10% validation subset was taken only from training. The custom Dataset parses year and state from each filename and gender from its folder. An undocumented integer follows paths in female split files; it was detected and ignored by taking the first whitespace-delimited field. The audit found zero missing or invalid listed images. To respect ACES inode quotas, only listed images were stored as blobs in one SQLite file, without changing examples or splits.

Images are decoded as RGB, resized/cropped to 224×224, and ImageNet-normalized. Training uses random resized crops, horizontal flips, and small affine shifts. Year is normalized to [0,1] with (year−1905)/108 and converted back to years before MAE calculation.

3. Models and training

Both systems use the same ImageNet-pretrained ResNet-50 backbone. The single-task head is dropout plus one linear output. MTL adds a two-logit gender head to the same shared representation. Both use normalized L1 year loss, AdamW (learning rate 3e-4, weight decay 1e-4), cosine decay, batch size 128, mixed precision, 20 maximum epochs, and identical preprocessing. MTL minimizes $L_{year} + \lambda L_{gender}$.

4. Experiments and validation-only selection

All random generators used seed 1. Every epoch saved model, optimizer, scheduler, scaler, epoch, and history for exact resume. Three MTL weights were trained; model and weight selection used validation only. The test loader was constructed only by the guarded final-evaluation command after selection.

Gender weight	Val year MAE	Val gender accuracy
0.05	2.550	98.97%
0.10	2.667	98.94%
0.20	2.702	98.87%

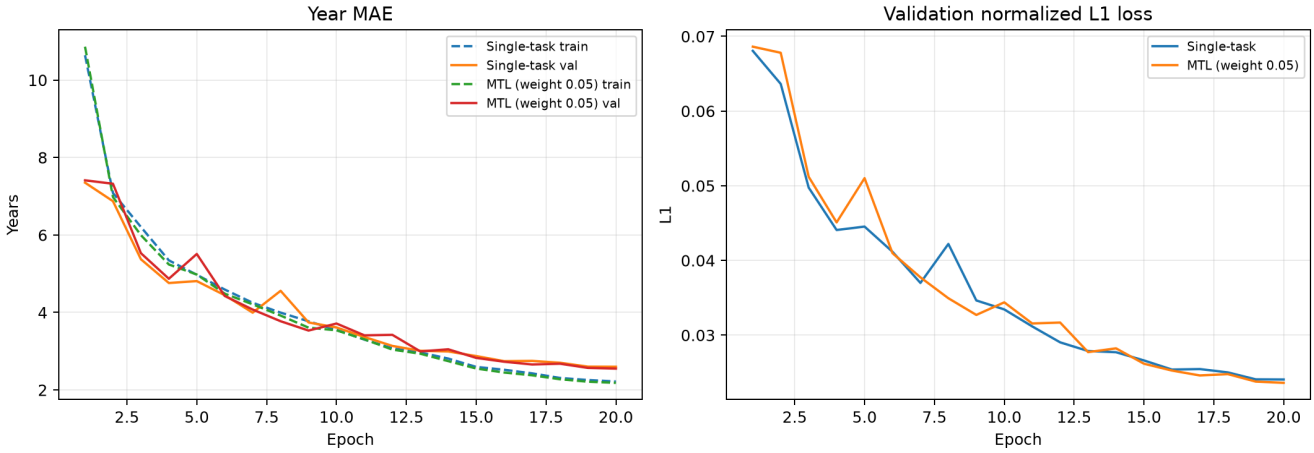


Figure 1. Year-regression learning curves for the single-task model and the validation-selected MTL model (λ=0.05).

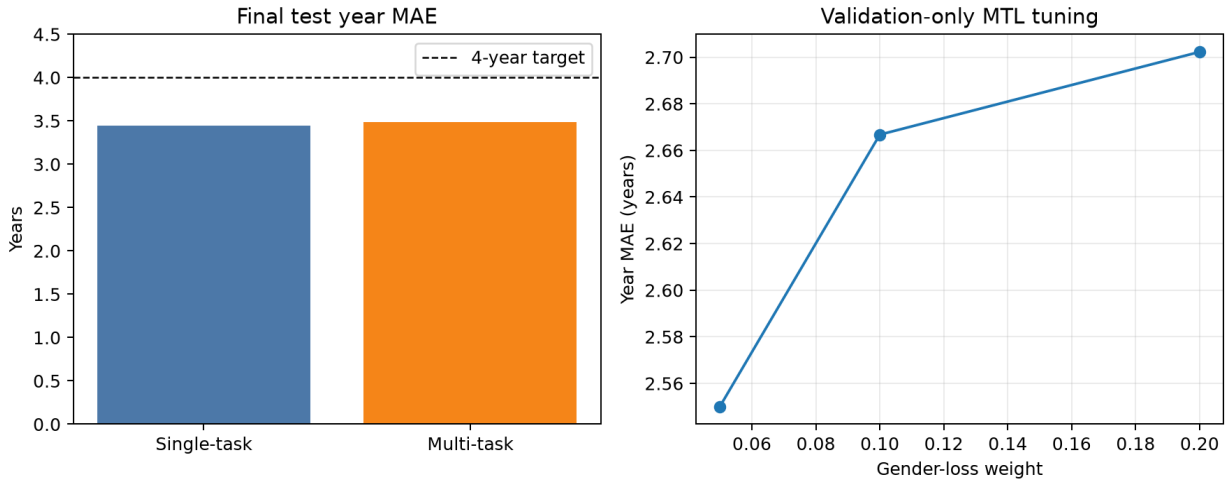


Figure 2. One-time final test MAE and validation-only auxiliary-weight tuning.

5. Final results and comparison

Model	Test year MAE	Gender accuracy / F1
Single-task	3.442 years	—
Multi-task (λ=0.05)	3.482 years	96.21% / 95.72%

The single-task model achieved the better test year result (3.442 years), while MTL reached 3.482 years. Both are below the four-year target. MTL improved validation MAE from 2.599 to 2.550 years, but that small gain did not transfer to the held-out test split: test MAE worsened by 0.040 years. This is evidence that the auxiliary task acted as a mild validation regularizer, not a reliable test improvement.

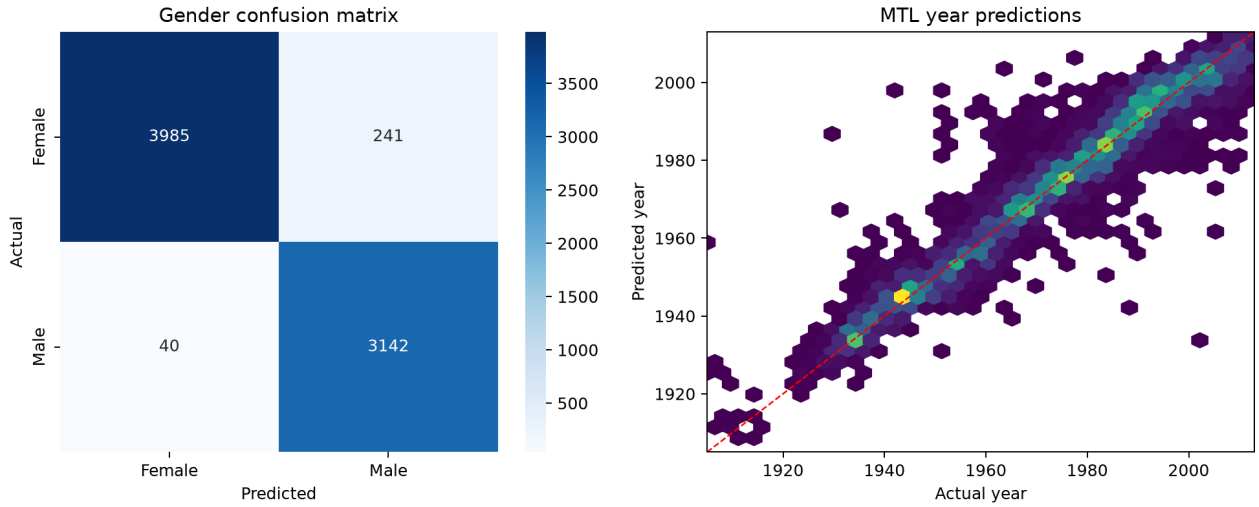


Figure 3. Gender confusion matrix and MTL year prediction density.

6. Discussion

Gender accuracy was 98.97% on validation but 96.21% on test, below the stated 97.5% expectation. The gap likely reflects temporal/school distribution shift between official lists and strong correlation between portrait style and year. The result was retained as-is: no checkpoint, weight, or epoch was selected using test performance. Other challenges were undocumented female-list suffixes, scratch inode limits, and unavailable PVC runtime initialization; compact SQLite storage and H100 execution resolved the infrastructure issues.

Potential improvements include stronger face-aligned augmentations, class-balanced gender sampling, calibration across decades, a larger modern backbone, ordinal or distributional age regression, and uncertainty-based dynamic task weighting. Any such change must be chosen on validation and followed by a new pre-registered final test run, rather than reusing this test result.